PATENT SPECIFICATION

Method and System for Detecting Hidden Risk Concentrations in Financial Networks Using Tensor Decomposition

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1. TECHNICAL FIELD

The present invention relates to the field of machine learning and anomaly detection, and more particularly to methods and systems for diagnosing and correcting structural failure modes in deployed classification models, with application to fraud detection in financial, aerospace, and medical domains.

2. BACKGROUND OF THE INVENTION

2.1. Technical Problem

Machine learning models deployed for fraud detection inevitably fail to detect certain classes of fraudulent activity. These failures are structural in nature—they arise from specific, identifiable deficiencies in how the model represents the data space—rather than from random noise. In production environments, these failures manifest as false negatives: fraudulent transactions that the model classifies as legitimate.

Current approaches to addressing model failures include:

(a) **Model retraining:** Collecting new labelled data and retraining the model from scratch. This is expensive, requires fresh labelled data (which may not be available), and may disrupt certified production systems that have undergone regulatory approval.

(b) **Ensemble methods:** Combining multiple models to reduce variance. While effective at reducing random errors, ensembles do not diagnose *why* specific instances are missed, and multiple models may share the same structural blind spot.

(c) **Cost-sensitive reweighting:** Adjusting class weights or decision thresholds. This is a global adjustment that trades precision for recall uniformly, without distinguishing between different failure modes.

(d) **Bias-variance decomposition:** The classical decomposition operates in *error space* (aggregate model performance), not in *feature space* (why a specific instance was missed). It provides no per-instance diagnostic information.

None of these approaches provides a structured, per-instance diagnosis of *why* a particular observation was missed, nor a targeted correction mechanism that addresses specific failure modes without modifying the underlying model.

2.2. Prior Art Limitations

Known prior art techniques for post-hoc model correction include conformal prediction (which provides calibrated prediction sets but does not diagnose failure modes), Bayesian uncertainty estimation (which quantifies overall prediction uncertainty but does not decompose it into actionable components), and SHAP/LIME-based explanation methods (which explain individual predictions but do not identify structural failure patterns or provide correction mechanisms).

3. SUMMARY OF THE INVENTION

The present invention provides a method and system for decomposing machine learning model failures into four near-orthogonal, measurable components that operate in feature space, and for applying targeted correction operators that recover missed detections without modifying the underlying model. The invention addresses the limitations of the prior art through the following technical contributions:

(i) A four-component decomposition of model failure modes into Camouflage (C), Feature Gap (G), Activity Anomaly (A), and Temporal Novelty (T), each computed from the feature space of the observation and reference distributions.

(ii) A self-calibrating weighting mechanism that computes component weights using one or more of Mutual Information, Fisher Variance Ratio, Bayesian Online, or Gradient-based methods.

(iii) A composite blind-spot score that aggregates the four components into a single scalar indicator of model vulnerability.

(iv) Post-hoc correction operators—including additive correction, logistic regression, quadratic surface fitting, and exponential gating—that apply targeted adjustments to model predictions based on the decomposed failure profile.

(v) Verification of near-orthogonality of the four components by normalised mutual information (mean NMI < 0.065) and variance inflation factor (all VIF < 1.45), confirming that each component captures an independent aspect of model failure.

4. DETAILED DESCRIPTION OF THE INVENTION

4.1. Four-Component Decomposition

The method operates on a frozen (unmodified) machine learning model f that produces predicted probabilities $p(x) = f(x)$ for input observations x. The method computes four components for each observation, each measuring a distinct structural aspect of why the model may fail on that observation.

4.1.1 Camouflage Component (C)

The camouflage component measures how closely a potentially fraudulent observation resembles the legitimate data distribution. It is defined as:

$$C(x) = 1 - \frac{\|x - \mu_{\text{legit}}\|_2}{d_{\max}} \tag{1}$$

where μ_{legit} is the mean of the legitimate class in the training data and $d_{\max}$ is the maximum distance from μ_{legit} observed in the training data. A value of $C(x)$ near 1 indicates high camouflage—the observation closely mimics legitimate patterns, making it difficult for the model to distinguish from genuine transactions.

4.1.2 Feature Gap Component (G)

The feature gap component measures the proportion of features that carry negligible information for the observation:

$$G(x) = \frac{|\{j : |x_j| < \varepsilon\}|}{d} \tag{2}$$

where d is the total number of features and $\varepsilon = 10^{-8}$ is a near-zero threshold. A high value of $G(x)$ indicates that many features are uninformative for this observation, depriving the model of the discriminative signal it relies upon.

4.1.3 Activity Anomaly Component (A)

The activity anomaly component measures whether the transactional activity profile of the observation deviates from the activity profiles of previously detected fraud:

$$A(x) = \sigma\left(\frac{\log(1 + |\text{tx\_count}(x)|) - \mu_{\text{caught}}}{\sigma_{\text{caught}}}\right) \tag{3}$$

where $\text{tx\_count}(x)$ is the transaction count associated with the observation, μ_{caught} and σ_{caught} are the mean and standard deviation of the log-transformed transaction counts of previously detected ("caught") fraudulent entities, and σ is the logistic sigmoid function. A value near 1 indicates that the observation's activity level is anomalous relative to known fraud patterns.

This component has no direct analogue in the classical bias-variance decomposition. It captures a distinct axis of model vulnerability: fraud operating at activity scales the model has not learned to associate with fraud.

4.1.4 Temporal Novelty Component (T)

The temporal novelty component measures the degree to which the observation represents a pattern that is temporally novel relative to the training data:

$$T(x) = \sigma\left(\frac{1}{2}(\bar{m}(x) - 2)\right) \tag{4}$$

where $\bar{m}(x)$ is the mean distance to the k nearest temporal neighbours in the training set, normalised by the temporal span. A value near 1 indicates high temporal novelty—a pattern that emerged after training and that the model cannot be expected to recognise.

4.2. Self-Calibrating Weight Computation

The four component values for an observation are aggregated into a composite blind-spot score (designated MFLS — Model Failure Likelihood Score) using weights that are computed automatically from the data:

$$\mathrm{MFLS}(x) = \sum_{i \in \{C,G,A,T\}} w_i \cdot S_i(x) \tag{5}$$

Four methods for computing the weights w_i are provided:

(a) **Mutual Information:** Weights are proportional to the mutual information between each component and the binary label (fraud vs. legitimate) on a held-out calibration set.

(b) **Fisher Variance Ratio:** Weights are proportional to the ratio of between-class variance to within-class variance for each component.

(c) **Bayesian Online:** Weights are updated sequentially using a Bayesian update rule as new labelled observations are received.

(d) **Gradient-based:** Weights are optimised to maximise detection performance on a holdout set via gradient descent.

The weights are normalised such that $\sum_i w_i = 1$.

4.3. Correction Operators

Given the four-component decomposition, the method applies correction operators to the frozen model's predicted probabilities to recover missed detections. Three correction operators are provided:

4.3.1 Additive Correction

The simplest correction adds a scaled composite score to the predicted probability for observations below a threshold:

$$p^*(x) = p(x) + \lambda \cdot \mathrm{MFLS}(x) \cdot (1 - p(x)) \cdot \mathbb{1}[p(x) < \tau] \tag{6}$$

where λ is a correction strength parameter and τ is a threshold below which correction is applied.

4.3.2 Logistic Regression Correction (SignedLR)

A lightweight logistic regression is trained on the four BSDT components as input features, predicting the true label. The correction is:

$$p^*(x) = \mathrm{clip}\left(p(x) + \beta_0 + \sum_i \beta_i S_i(x),\ 0,\ 1\right) \tag{7}$$

4.3.3 Quadratic Surface Correction (QuadSurf)

A degree-2 polynomial ridge regression is fitted on the four components and their interactions, providing a richer correction surface:

$$p^*(x) = \text{clip}\left(p(x) + \phi_2([C, G, A, T])^\top \hat{\beta},\ 0,\ 1\right) \tag{8}$$

where ϕ_2 denotes the degree-2 polynomial feature map that includes all terms S_i, S_i^2, and $S_i S_j$ for $i \neq j$. In the preferred embodiment, ridge regularisation with coefficient $\alpha = 1.0$ is applied.

4.3.4 Exponential Gating Correction (ExpGate)

A gated correction combines a quadratic surface with a sigmoid-activated gate:

$$p^*(x) = \text{clip}\left(p(x) + g(x) \cdot q(x),\ 0,\ 1\right) \tag{9}$$

where $q(x)$ is the quadratic surface output and $g(x) = \sigma(v^\top [C, G, A, T] + b)$ is a learned sigmoid gate that modulates the correction strength.

4.4. Seven Supplementary Features

In an enhanced embodiment, seven supplementary features are derived from the core decomposition for use by the correction operators:

- φ_1–φ_4: The four raw components C, G, A, T.
- φ_5: The MFLS composite score.
- φ_6: The predicted-vs-actual residual: $|p(x) - y|$ where y is the true label (available only in supervised calibration).
- φ_7: An interaction term: $C(x) \cdot A(x)$, capturing the joint effect of high camouflage with anomalous activity.

4.5. Orthogonality Verification

The four components are verified to be near-orthogonal by three independent measures:

- **Normalised Mutual Information:** Mean NMI across all component pairs ranges from 0.051 to 0.064, well below the redundancy threshold of 0.30.
- **Variance Inflation Factor:** All VIF values are below 1.45, confirming negligible multicollinearity.
- **Principal Component Analysis:** All four components carry meaningful variance; no component is dominated by others.

4.6. Cross-Domain Applicability

The method has been validated across seven datasets spanning four domains:

Domain	Dataset	Dominant Component
Blockchain	Elliptic	Temporal Novelty (T)
Blockchain	Ethereum Addresses	Temporal Novelty (T)
Banking	Credit Card Fraud	Activity Anomaly (A)
Banking	IEEE-CIS	Activity Anomaly (A)
Aerospace	Shuttle	Camouflage (C)
Medical	Mammography	Camouflage (C)
Pattern Recognition	Pendigits	Feature Gap (G)

The domain-dependent variation in dominant components confirms that the decomposition captures genuine structural differences in failure modes across application domains.

4.7. Performance Characteristics

In the preferred embodiment, evaluated across 792 variant–model–dataset combinations:

Metric	Value
Best variant (QuadSurf) mean F1	0.787 (+43% relative improvement)
False positive reduction (QuadSurf)	88.1% (44,740 $\rightarrow$ 5,345)
Maximum recall recovery	+84.4 percentage points
Shuttle domain rescue: F1	0.134 $\rightarrow$ 0.968
Shuttle domain FP reduction	99.94%
Zero-missed-fraud configurations	10/12 model–dataset combos
Cross-domain credit card MFLS AUC	0.885
Cross-domain aerospace MFLS AUC	0.982
Cross-domain medical imaging MFLS AUC	0.916

4.8. Cross-Domain Transfer

The damping mechanism derived from the BSDT framework transfers to physical systems. Specifically, the adaptive damping function:

$$\gamma^*(E) = \frac{E}{E + \theta} \tag{10}$$

where E is the blind-spot energy and θ is a damping threshold, has been validated as a viscosity regularisation term in Navier–Stokes fluid dynamics simulations, representing a novel finance-to-physics transfer direction.

5. CLAIMS

The following claims define the scope of protection sought. Where a claim refers to another claim, the dependency is by claim number. Features of any dependent claim may be combined with any independent claim or with other dependent claims.

1. A computer-implemented method for detecting and correcting failure modes in a deployed machine learning classification model without modifying the model, the method comprising the steps of:

(a) receiving a feature vector x and a predicted probability $p(x)$ output by the deployed classification model;

(b) computing a camouflage component $C(x)$ that measures the proximity of the feature vector to the centroid of a reference legitimate population in feature space;

(c) computing a feature gap component $G(x)$ that measures the proportion of features in the feature vector having values below a near-zero threshold;

(d) computing an activity anomaly component $A(x)$ that measures the deviation of a transactional activity metric of the observation from the activity profile of a reference population of previously detected anomalies;

(e) computing a temporal novelty component $T(x)$ that measures the degree to which the observation is temporally distant from patterns present in the training data;

(f) computing a composite blind-spot score by aggregating the four components using automatically calibrated weights; and

(g) applying a correction operator to the predicted probability $p(x)$ based on values of one or more of the computed components to produce a corrected probability $p^*(x)$.

2. A system for detecting structural failure modes in a deployed machine learning classification model, the system comprising:

(a) a decomposition module configured to compute, for each input observation, four near-orthogonal failure mode components comprising:

  (i) a camouflage component measuring proximity to a legitimate-class centroid;

  (ii) a feature gap component measuring informational sparsity;

  (iii) an activity anomaly component measuring deviation from detected-anomaly activity patterns; and

  (iv) a temporal novelty component measuring temporal distance from training data patterns;

(b) a weighting module configured to compute component weights by one or more of mutual information, Fisher variance ratio, Bayesian online updating, or gradient-based optimisation;

(c) an aggregation module configured to compute a composite blind-spot score from the weighted components; and

(d) a correction module configured to apply one or more correction operators to predicted probabilities output by the deployed model, based on the computed component values, to produce corrected probabilities;

wherein the system operates on a frozen deployed model without requiring modification or retraining of the deployed model.

3. A method according to claim 1, wherein the camouflage component is computed as:

$$C(x) = 1 - \frac{\|x - \mu_{\text{legit}}\|_2}{d_{\max}}$$

where μ_{legit} is the centroid of the legitimate class and $d_{\max}$ is the maximum observed distance from the centroid.

4. A method according to claim 1, wherein the feature gap component is computed as:

$$G(x) = \frac{|\{j : |x_j| < \varepsilon\}|}{d}$$

where d is the number of features and ε is a near-zero threshold.

5. A method according to claim 1, wherein the activity anomaly component is computed as:

$$A(x) = \sigma\left(\frac{\log(1 + |\text{activity}(x)|) - \mu_{\text{ref}}}{\sigma_{\text{ref}}}\right)$$

where $\text{activity}(x)$ is a transactional activity metric, μ_{ref} and σ_{ref} are statistics of a reference detected-anomaly population, and σ is the logistic sigmoid function.

6. A method according to claim 1, wherein the correction operator is a quadratic surface correction comprising:

$$p^*(x) = \text{clip}\left(p(x) + \phi_2([C, G, A, T])^\top \hat{\beta},\ 0,\ 1\right)$$

where ϕ_2 is a degree-2 polynomial feature map of the four components and $\hat{\beta}$ are coefficients fitted by ridge regression.

7. A method according to claim 1, wherein the correction operator is an additive correction comprising:

$$p^*(x) = p(x) + \lambda \cdot \text{MFLS}(x) \cdot (1 - p(x)) \cdot \mathbb{1}[p(x) < \tau]$$

where λ is a correction strength parameter and τ is a threshold.

8. A method according to claim 1, wherein the correction operator is an exponential gating correction comprising a quadratic surface output modulated by a learned sigmoid gate.

9. A method or system according to any preceding claim, wherein the four components are verified to be near-orthogonal by having normalised mutual information below 0.30 for all pairs and variance inflation factors below 2.0 for all components.

10. A method or system according to any preceding claim, wherein the dominant component varies by application domain, with temporal novelty dominating in blockchain domains, activity anomaly dominating in banking domains, and camouflage dominating in aerospace and medical domains.

11. A method or system according to any preceding claim, further comprising the step of computing seven supplementary features derived from the four components, comprising the four raw components, the composite score, a predicted-vs-actual residual, and an interaction term between the camouflage and activity anomaly components.

12. A method or system according to any preceding claim, wherein the adaptive damping function $\gamma^*(E) = E/(E + \theta)$ derived from the composite blind-spot energy is further applied as a viscosity regularisation term in a computational fluid dynamics simulation.

6. DRAWINGS

[To be filed separately: Diagram of the four-component decomposition; flow diagram of the correction operator pipeline; orthogonality verification plots (NMI matrix, VIF chart); cross-domain performance comparison chart; schematic of the BSDT layer operating on a frozen model.]

ABSTRACT

A computer-implemented method and system detect and correct structural failure modes in deployed machine learning classification models without model retraining. Model failures are decomposed into four near-orthogonal components: a camouflage component measuring similarity of a fraudulent observation to legitimate data; a feature gap component measuring informational sparsity; an activity anomaly component measuring deviation from established fraud detection patterns; and a temporal novelty component measuring novelty relative to the training distribution. A composite score aggregates the four components using self-calibrating weights. Post-hoc correction operators apply targeted adjustments to predicted probabilities based on the decomposed failure profile, recovering missed detections at improvements of up to 84.4 percentage points of recall. The method operates on any classification model without modifying its parameters, provides interpretable attribution of failure cause, and is validated across financial, aerospace and medical datasets.